

Research

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Coloring Big Graphs With AlphaGoZero

Link: <https://arxiv.org/pdf/1902.10162>

Summary: This project adapts the AlphaGoZero reinforcement learning algorithm by integrating it with graph embeddings to learn fast and effective heuristics for graph coloring entirely from scratch

Solution & Method: The authors formulate graph coloring as a Markov Decision Process (MDP). The method utilizes deep reinforcement learning alongside Monte Carlo Tree Search (MCTS) with Upper Confidence Bound (UCB) and self-play. To accommodate varied graph sizes, they introduce FastColorNet, a deep neural network architecture that computes a dynamically sized softmax assignment to predict the best coloring strategies by accessing the full global graph context

Benchmarks: The models were trained on synthetic random graphs and benchmarked on real-world graphs sourced from the publicly available SuiteSparse matrix collection. The sources do not specify a download link for their proprietary training framework.

Other Info: AlphaGo Zero is a groundbreaking DeepMind AI that mastered the game of Go in just three days using only reinforcement learning and self-play, without human data. Its key innovation is learning solely from the raw board state, abandoning human-precedent for a more efficient, superhuman approach. Reinforcement learning (RL) is a machine learning method where an autonomous agent learns to make optimal, sequential decisions by interacting with an environment through trial and error.

Results:

Dataset	ER-1K	WS-1K	ER-16K	WS-16K	ER-10M	WS-10M	SS-CIR	SS-LP	SS-Web	SS-FE
Unordered	34.3	59.2	732.8	265.35	42923	16415	4.2	4.25	3.75	4.85
Ordered	32.45	57.35	715.2	261.8	40347	15922	3.15	2.95	2.6	4.05
Dynamic	32.2	57.15	708.5	261.2	37524	15843	3.55	3.15	2.7	4.25
FCN-train	29.58	52.5	660.19	237.03	35362	14924	3.0	2.95	2.4	3.75
FCN-test	31.7	56.59	702.57	258.3	37849	15023	3.1	2.95	2.55	4.1
FCN-gen	33.9	57.66	708.13	267.53	43415	17262	4.15	4.3	3.7	4.95

Graph Colouring Meets Deep Learning: Effective Graph Neural Network Models for Combinatorial Problems

Link: <https://arxiv.org/pdf/1903.04598>

Summary: This article introduces a Graph Neural Network (GNN) framework aimed at solving the decision version of the graph coloring problem (GCP) without requiring prior reductions to other problems like SAT

Solution & Method: The solution relies on a message-passing algorithm where two types of nodes, the graph’s vertices and the candidate colors, communicate to update their multidimensional embeddings. Trained as a binary classifier to answer whether a graph accepts a specific coloring, it uses an adversarial training strategy utilizing graphs generated on the verge of satisfiability (phase transition). Post-processing extracts constructive color assignments by running a k-means clustering algorithm on the vertex embeddings

Benchmarks: Tested on standard instances from the COLOR02/03/04 Workshop dataset and various random graphs (power-law trees, small-world networks). The models and instances: <https://machine-reasoning-ufrgs.github.io/GNN-GCP/>

Other Info: Message passing in graphs entails nodes exchanging information (messages) with their neighbors iteratively. This process updates each node’s state based on aggregated information, enabling nodes to integrate knowledge from their local neighborhoods. This correlation in graph networks supports tasks such as node classification or link prediction.

Results:

Instance	Size	χ	Computed χ		
			GNN	Tabucol	Greedy
queen5_5	25	5	6	5	8
queen6_6	36	7	7	8	11
myciel5	47	6	5	6	6
queen7_7	49	7	8	8	10
queen8_8	64	9	8	10	13
1-Insertions_4	67	4	4	5	5
huck	74	11	8	11	11
jean	80	10	7	10	10
queen9_9	81	10	9	11	16
david	87	11	9	11	12
mug88_1	88	4	3	4	4
myciel6	95	7	7	7	7
queen8_12	96	12	10	12	15
games120	120	9	6	9	9
queen11_11	121	11	12	NA	17
anna	138	11	11	11	12
2-Insertions_4	149	4	4	5	5
queen13_13	169	13	14	NA	21
myciel7	191	8	NA	8	8
homer	561	13	14	13	15

Graph Coloring with Physics-Inspired Graph Neural Networks

Link: <https://arxiv.org/pdf/2202.01606>

Summary: This project utilizes concepts from statistical physics to solve graph coloring as an unsupervised multi-class node classification problem via GNNs

Solution & Method: The method strictly relies on an unsupervised training strategy directed by a differentiable loss function based on the anti-ferromagnetic Potts model, a statistical physics concept that naturally applies energy penalties when adjacent nodes share the same color. The GNN (using structures like GCN or GraphSAGE) outputs a soft, probabilistic distribution of color classes, which are then mapped to distinct color assignments using a simple projection heuristic (like argmax)

Benchmarks: Benchmarked on standard instances from the COLOR dataset (like Queens and Myciel graphs) and large real-world citation graphs (Cora, Citeseer, Pubmed). A demo: <https://github.com/amazon-research/gcp-with-gnns-example>

Other Info: -

Results:

graph	nodes	edges	density	colors q	$\bar{\chi}_{\text{greedy}}$	$\bar{\chi}_{\text{GNN}}$	Tabucol	Tabucol [33]	GNN [33]	PI-GCN	PI-SAGE	error ϵ
anna	138	493	5.22%	11	11	11	0	0	1	1	0	0.00%
jean	80	254	8.04%	10	10	10	0	0	0	0	0	0.00%
myciel5	47	236	21.83%	6	6	6	0	0	0	0	0	0.00%
myciel6	95	755	16.91%	7	7	7	0	0	0	0	0	0.00%
queen5-5	25	160	53.33%	5	5	5	0	0	0	0	0	0.00%
queen6-6	36	290	46.03%	7	8	7	0	0	4	1	0	0.00%
queen7-7	49	476	40.48%	7	9	7	0	10	15	8	0	0.00%
queen8-8	64	728	36.11%	9	10	10	0	8	7	6	1	0.14%
queen9-9	81	1056	32.59%	10	12	11	0	5	13	13	1	0.09%
queen8-12	96	1368	30.00%	12	13	12	0	10	7	10	0	0.00%
queen11-11	121	1980	27.27%	11	15	14	20	33	33	37	17	0.86%
queen13-13	169	3328	23.44%	13	17	17	35	42	40	61	26	0.78%

Rethinking Graph Neural Networks for the Graph Coloring Problem

Link: <https://arxiv.org/pdf/2208.06975>

Summary: The article diagnoses why popular aggregation-combine GNNs (AC-GNNs) often fail at graph coloring and proposes a simple GNN architecture designed to overcome these mathematical limitations

Solution & Method: They propose the Graph Discrimination Network (GDN), which avoids combining the node's ego-information with its neighbors, uses an unsupervised margin loss to ensure connected nodes receive distinct probability distributions, and incorporates customized tensor-based preprocess and postprocess operations. It also proposes a combination with Integer Linear Programming (ILP) solvers to further refine complex results

Benchmarks: Evaluated on a real-world Layout dataset from circuit design, Citation datasets (Cora, Citeseer, Pubmed), and the COLOR dataset

Other Info: Integer Linear Programming (ILP) is an optimization technique that deals with problems involving linear relationships and integer variables. It extends Linear Programming (LP) by restricting decision variables to integer values. (facut si la AEA, curs+seminar 2)

Results:

Graph	$ \mathcal{V} $	$ \mathcal{E} $	$d\%$	$\mathcal{X}(k)$	GNN-GCP		Tabucol		HybridEA		GDN	
					Cost	Time	Cost	Time	Cost	Time	Cost	Time
35158	641202	787242	-	3	386009	3896	2392	82301	1562	133285	1557 ±45	5.84 ±0.23
					247.9	667.1	1.54	14092	1.00	22822	1.0	1.0
Cora	2708	5429	0.15	5	1291	3.90	31	15410	0	18921	0 ±0	0.81 ±0.08
Citeseer	3327	4732	0.09	6	1733	2.74	6	44700	0	24230	0 ±0	1.42 ±0.15
Pubmed	19717	44338	0.03	8	4393	4.50	-	>24h	-	>24h	21 ±4	1.41 ±0.12
					353.2	3.06	-	-	-	-	1.0	1.0
jean	80	254	8	10	76	0.06	0	0.95	0	0.01	0 ±1	0.13 ±0.02
anna	138	493	5	11	87	0.08	0	3.23	0	0.02	0 ±0	0.17 ±0.03
huck	74	301	11	11	117	0.05	0	0.15	0	0.01	0 ±0	0.06 ±0.02
david	87	406	11	11	-	-	0	4.83	0	0.01	0 ±0	0.19 ±0.01
homer	561	1628	1	13	1628	1.09	0	274	0	0.06	0 ±1	0.29 ±0.02
myciel5	47	236	22	6	35	0.04	0	0.20	0	0.01	0 ±0	0.12 ±0.01
myciel6	95	755	17	7	94	4.33	0	0.79	0	0.01	0 ±0	0.21 ±0.02
games120	120	638	9	9	301	0.07	0	0.93	0	0.01	0 ±1	0.08 ±0.01
Mug88_1	88	146	4	3	146	0.33	0	0.12	0	0.01	0 ±0	0.01 ±0
1-Insertions_4	67	232	10	2	42	0.05	0	0.16	0	0.01	0 ±0	0.07 ±0
2-Insertions_4	212	1621	7	4	360	0.09	1	255	1	101.1	1 ±0	60.08 ±0.01
Queen5_5	25	160	53	5	37	0.03	0	0.13	0	0.07	0 ±0	0.05 ±0.01
Queen6_6	36	290	46	6	290	0.38	0	4.93	0	1.89	0 ±0	0.08 ±0
Queen7_7	49	476	40	7	126	0.04	10	36.9	9	51.8	9 ±1	0.38 ±0.08
Queen8_8	64	728	36	8	188	0.05	8	61.3	5	74.1	2 ±0	0.13 ±0.03
Queen9_9	81	1056	33	9	296	0.07	5	97.8	6	126.9	6 ±1	0.09 ±0.01
Queen8_12	96	1368	30	12	260	0.10	10	139	3	92.9	0 ±0	0.58 ±0.09
Queen11_11	121	3960	55	11	396	0.10	33	213	22	141.3	21 ±3	0.07 ±0.01
Queen13_13	169	6656	47	13	728	0.20	42	401	37	213	33 ±2	2.38 ±0.32
					-	-	1.51	22.63	1.15	12.10	1.0	1.0

Generating a Graph Colouring Heuristic with Deep Q-Learning and Graph Neural Networks

Link: <https://arxiv.org/pdf/2304.04051>

Summary: This project focuses on automatically generating competitive greedy construction heuristics for coloring by framing node selection as a sequential decision-making task

Solution & Method: The ReLCol algorithm integrates Deep Q-learning (DQN) with a GNN to sequentially select the best next uncolored vertex. A major innovation is the parameterization of the state as a complete graph; this structural alteration allows the GNN’s message-passing mechanism to freely exchange data between all pairs of vertices, not exclusively adjacent ones

Benchmarks: The models were trained on randomly generated graphs (including Erdős-Renyi and Barabasi-Albert networks). The tests were performed against the COLOR02 benchmark dataset and Spinrad graphs. The generating algorithms and datasets are available at https://github.com/gpdwatkins/graph_colouring_with_RL

Other Info: -

Results:

Graph instance	n	χ	Random	LF	SL	DSATUR	Gianinazzi et al.	ReLCol
queen5_5	25	5	$2.3^{\pm 0.1}$	2	3	0	$2.1^{\pm 0.3}$	$0.2^{\pm 0.11}$
queen6_6	36	7	$2.4^{\pm 0.06}$	2	4	2	$3.2^{\pm 0.16}$	$1^{\pm 0}$
myciel5	47	6	$0.1^{\pm 0.03}$	0	0	0	$0.1^{\pm 0.08}$	$0^{\pm 0}$
queen7_7	49	7	$4^{\pm 0.06}$	5	3	4	$3.3^{\pm 0.32}$	$2.2^{\pm 0.16}$
queen8_8	64	9	$3.4^{\pm 0.07}$	4	5	3	$3.3^{\pm 0.24}$	$2.1^{\pm 0.14}$
1-Insertions_4	67	4	$1.2^{\pm 0.04}$	1	1	1	$1^{\pm 0}$	$1^{\pm 0}$
huck	74	11	$0^{\pm 0}$	0	0	0	$0^{\pm 0}$	$0^{\pm 0}$
jean	80	10	$0.3^{\pm 0.05}$	0	0	0	$0^{\pm 0}$	$0.3^{\pm 0.12}$
queen9_9	81	10	$3.9^{\pm 0.07}$	5	5	3	$5^{\pm 0}$	$2.7^{\pm 0.25}$
david	87	11	$0.7^{\pm 0.07}$	0	0	0	$0.4^{\pm 0.14}$	$1.2^{\pm 0.33}$
mug88_1	88	4	$0.1^{\pm 0.02}$	0	0	0	$0^{\pm 0}$	$0^{\pm 0}$
myciel6	95	7	$0.3^{\pm 0.05}$	0	0	0	$0.8^{\pm 0.17}$	$0^{\pm 0}$
queen8_12	96	12	$3.3^{\pm 0.06}$	3	3	2	$4^{\pm 0}$	$2.6^{\pm 0.18}$
games120	120	9	$0^{\pm 0.01}$	0	0	0	$0^{\pm 0}$	$0^{\pm 0}$
queen11_11	121	11	$5.9^{\pm 0.07}$	6	6	4	$6^{\pm 0}$	$5.4^{\pm 0.25}$
anna	138	11	$0.2^{\pm 0.04}$	0	0	0	$0^{\pm 0}$	$0.4^{\pm 0.18}$
2-Insertions_4	149	4	$1.5^{\pm 0.05}$	1	1	1	$1^{\pm 0}$	$1^{\pm 0}$
queen13_13	169	13	$6.5^{\pm 0.07}$	10	9	4	$8^{\pm 0}$	$6.3^{\pm 0.24}$
myciel7	191	8	$0.6^{\pm 0.06}$	0	0	0	$0.3^{\pm 0.14}$	$0.2^{\pm 0.11}$
homer	561	13	$1^{\pm 0.07}$	0	0	0	$0^{\pm 0}$	$0.6^{\pm 0.22}$
Average			$1.9^{\pm 0.01}$	1.95	2	1.2	$1.92^{\pm 0.05}$	$1.35^{\pm 0.05}$

Graph Neural Networks as Ordering Heuristics for Parallel Graph Coloring

Link: <https://arxiv.org/pdf/2408.05054>

Summary: This paper proposes a highly optimized, shallow GNN-based ordering heuristic built explicitly to compete with the fastest non-trivial greedy heuristics in execution speed, solution quality, and parallel scalability

Solution & Method: The method utilizes a very shallow GraphSAGE model (only 2 to 4 layers) to directly predict node priority values, constructing a partial ordering. This generated ordering is subsequently processed using the parallel Jones-Plassmann (JP) algorithm to color the graph concurrently. Training occurs in two stages: supervised learning on fast established heuristics (like Smallest Degree Last), followed by population-based genetic training to directly mutate model parameters and refine coloring numbers

Benchmarks: Evaluated on 67 graphs from the Stanford Large Network Dataset Collection (SNAP) and 80 instances from the DIMACS implementation challenge. Both the source code (written in highly optimized C) and trained models are open-source at <https://github.com/KennethLangedal/GNN-COLORING>

Other Info: Ordering heuristics are rules of thumb used in algorithms, particularly backtracking searches for Constraint Satisfaction Problems (CSPs), to determine the sequence in which variables are assigned or values are tried. They aim to minimize search space and find solutions faster by prioritizing choices that are most likely to fail early ("first-fail") or by utilizing domain-specific information, such as the Least Constrained Variable or Minimum Remaining Values (MRV).

Results: -

Solving the graph coloring problem via hybrid genetic algorithms

Link: <https://www.sciencedirect.com/science/article/pii/S1018363913000135>

Summary: The paper proposes a method that uses evolutionary search (genetic algorithms) to explore possible colorings and a local optimization heuristic (DBG) to fix conflicts, producing efficient solutions for the graph coloring problem.

Solution & Method: The genetic algorithm (GA) explores the solution space using evolutionary principles. The GA performs global exploration of the search space. To improve solutions, the algorithm integrates a local search heuristic called DBG. The DBG procedure: examines vertices causing conflicts, attempts to reassign colors locally, reduces the number of color conflicts, moves the solution toward a valid coloring. This step performs local optimization. This hybridization allows the method to: escape local minima, improve solution quality, converge faster.

Benchmarks: The algorithm is tested on DIMACS benchmark graphs, and the results show that the proposed hybrid method produces competitive and effective solutions compared with other algorithms for graph coloring.(Graph Coloring Instances)

Other Info: -

Results:

Table 1 Experimental results. ⋮						
Graph	$ V $	$ E $	k^*	k_{Ours}	$T_{avg}(s)$	succ
anna	138	493	11	11	11.63	12/15
david	87	406	11	11	10.89	14/15
huck	74	301	11	11	11.10	15/15
2-Inser-3	37	72	4	4	2.23	15/15
3-Inser-3	56	110	4	4	3.37	15/15
jean	80	254	10	10	9.92	15/15
queen5.5	25	160	5	5	1.20	15/15
queen6.6	36	290	7	7	1.32	15/15
queen7.7	49	476	7	7	6.91	15/15
queen8.8	64	728	9	9	9.87	11/15
queen9.9	81	1056	10	10	13.96	13/15
miles250	128	387	8	8	4.39	11/15
miles500	128	1170	20	20	14.48	10/15
games 120	120	638	9	9	10.77	15/15
mug88-1	88	146	4	4	4.05	15/15
mug88-25	88	146	4	4	3.67	13/15
mug 100-1	100	166	4	4	8.34	11/15
mug 100-25	100	166	4	4	8.21	13/15
myciel3	11	20	4	4	0.01	15/15
myciel4	23	71	5	5	0.89	15/15
myciel5	47	236	6	6	5.41	15/15
myciel6	95	755	7	7	12.77	11/15
myciel7	191	2360	8	8	20.19	9/15
dsjc 125.1	125	736	5	6	13.12	14/15
dsjcl 125.5	125	3891	17	17	19.71	8/15
dsjcl 125.9	125	6961	44	44	35.87	7/15
dsjcl250.1	250	3218	8	8	26.07	10/15
dsjcl250.9	250	27897	72	72	75.81	6/15
fpsol2.i.1	496	11654	65	65	82.03	2/15
fpsol2.i.2	451	8691	30	30	69.79	6/15
fpsol2.i.3	425	8688	30	30	67.14	4/15
zeroin.i.1	211	4100	49	49	28.24	5/15
zeroin.i.2	211	3541	30	30	83.39	9/15
zeroin.i.3	206	3540	30	30	27.06	6/15
mulsol.i.1	197	3925	49	49	25.75	2/15
mulsol.i.2	188	3885	31	31	22.07	7/15
mulsol.i.3	184	3916	31	31	23.49	5/15
mulsol.i.4	185	3946	31	31	25.22	4/15
mulsol.i.5	186	3973	31	31	28.33	7/15

Graph	$ V $	Wu and Hao (2012)	Lu and Hao (2010)	Blochliher and Zufferey (2008)	Hertz et al. (2008)	Malaguti and Monaci (2008)	Chiarandini and Stutzle (2002)	k_{Ours}	$T_{avg}(s)$	succ
dsjc250.5	250	*	28	*	*	28	28	28	89	18/20
dsjc500.1	500	*	12	12	12	12	12	12	112	15/20
dsjc500.5	500	*	48	48	48	48	49	48	147	18/20
dsjc500.9	500	*	126	127	126	127	126	126	190	19/20
dsjc1000.1	1000	20	20	20	20	20	*	20	4982	9/20
dsjc1000.5	1000	83	83	89	86	83	89	83	2164	13/20
dsjc1000.9	1000	222	223	226	224	224	*	224	8092	7/20
dsjr500.1c	500	*	85	85	85	85	*	85	581	16/20
dsjr500.5	500	*	122	125	125	122	124	124	728	13/20
r250.5	250	*	65	66	*	65	*	65	261	17/20
r250.1	250	*	*	*	*	8	*	8	308	20/20
r250.1c	250	*	*	64	*	64	*	64	266	18/20
r1000.1	1000	*	*	*	*	20	*	20	895	10/20
r1000.1c	1000	101	98	98	*	98	*	98	1329	6/20
r1000.5	1000	249	245	248	*	234	*	242	1507	3/20
le450_15c	450	*	15	15	15	15	15	15	93	20/20
le450_15d	450	*	15	15	15	15	15	15	228	20/20
le450_25c	450	*	25	25	25	25	26	25	740	20/20
le450_25d	450	*	26	25	25	25	26	25	382	18/20
flat300_20_0	300	*	*	*	*	20	*	20	241	20/20
flat300_26_0	300	*	26	*	*	26	26	26	275	16/20
flat300_28_0	300	*	29	28	28	31	31	28	319	18/20
flat1000_50_0	1000	50	50	50	50	50	*	50	802	9/20
flat1000_60_0	1000	60	60	60	60	60	*	60	1344	5/20
flat1000_76_0	1000	82	82	87	85	82	*	82	3795	7/20
C2000.5	2000	146	148	*	*	*	*	151	8421	8/20
C2000.9	2000	409	*	*	*	*	*	411	7368	5/20
C4000.5	4000	260	272	*	*	*	*	282	212881	2/20
latin_sqr_10	900	*	99	*	*	101	99	99	1267	11/20

1 Benchmark instances

<https://mat.tepper.cmu.edu/COLOR/instances.html>

DSJ: random graphs and DSJC are standard (n,p) random graphs. DSJR are geometric graphs, with DSJR..c being complements of geometric graphs.

CUL: Quasi-random coloring problem.

REG: Problem based on register allocation for variables in real codes.

LEI: Leighton graphs with guaranteed coloring size.

SCH: Class scheduling graphs, with and without study halls.

LAT: Latin square problem.

SGB: Graphs from Donald Knuth's Stanford GraphBase. These can be divided into:

1. Book Graphs. Given a work of literature, a graph is created where each node represents a character. Two nodes are connected by an edge if the corresponding characters encounter each other in the book. Knuth creates the graphs for five classic works: Tolstoy's Anna Karenina (anna), Dicken's David Copperfield (david), Homer's Iliad (homer), Twain's Huckleberry Finn (huck), and Hugo's Les Misérables (jean).

2. Game Graphs. A graph representing the games played in a college football season can be represented by a graph where the nodes represent each college team. Two teams are connected by an edge if they played each other during the season. Knuth gives the graph for the 1990 college football season.

3. Miles Graphs. These graphs are similar to geometric graphs in that nodes are placed in space with two nodes connected if they are close enough. These graphs, however, are not random. The nodes represent a set of United States cities and the distance between them is given by road mileage from 1947. These graphs are also due to Kuth.

4. Queen Graphs. Given an n by n chessboard, a queen graph is a graph on n^2 nodes, each corresponding to a square of the board. Two nodes are connected by an edge if the corresponding squares are in the same row, column, or diagonal. Unlike some of the other graphs, the coloring problem on this graph has a natural interpretation: Given such a chessboard, is it possible to place n sets of n queens on the board so that no two queens of the same set are in the same row, column, or diagonal? The answer is yes if and only if the graph has coloring number n . Martin Gardner states without proof that this is the case if and only if n is not divisible by either 2 or 3. In all cases, the maximum clique in the graph is no more than n , and the coloring value is no less than n .

MYC: Graphs based on the Mycielski transformation. These graphs are difficult to solve because they are triangle free (clique number 2) but the coloring number increases in problem size.

COLOR02/03/04: Graph Coloring and its Generalizations <https://mat.tepper.cmu.edu/COLOR02/>

DIMACS Graphs: Benchmark Instances and Best Upper Bounds <https://cedric.cnam.fr/~porumbed/graphs/>

the result tables are now split, using a classification already proposed in certain papers:

- (i) local search algorithms (no populations, no hybridisations, no independent set isolation) and
- (ii) population-based algorithms, complex hybrid EAs, etc.

Graph	best K / χ	References	Graph	best K / χ	References	Graph	best K / χ	References
dsjc250.5	28/?	[2,3,7,15,18]	r250.5	66/65	[1]	flat300_28_0	28/28	[1,15,18]
dsjc500.1	12/?	[1,2,15,18]	r1000.1c	98/?	[1,18]	flat1000_50_0	50/50	[1,15]
dsjc500.5	48/?	[1,15,18]	r1000.5	234/234	[13]	flat1000_60_0	60/60	[1,15]
dsjc500.9	126/?	[1,15,18]	dsjr500.1c	84/84	[17]	flat1000_76_0	85/76	[18]
dsjc1000.1	20/?	[1,14,15,18]	dsjr500.5	122/122	[13]	latin_square	99/?	[2]
dsjc1000.5	85/?	[18]	le450_25c	25/25	[1,18]	C2000.5	/?	-
dsjc1000.9	223/?	[18]	le450.25d	25/25	[1,18]	C4000.5	/?	-

Hybrid and population-based methods								
Graph	best K / χ	References	Graph	best K / χ	References	Graph	best K / χ	References
dsjc250.5	28/?	[6,7,12,14,18,19,20,22]	r250.5	65/65	[12,14,19,20,22,23]	flat300_28_0	29/28	[19,20]
dsjc500.1	12/?	[7,11,12,14,19,20,22]	r1000.1c	98/?	[12,14,19,20,22]	flat1000_50_0	50/50	[12,7,14,19,20,21]
dsjc500.5	48/?	[6,7,11,14,19,20,22,23]	r1000.5	234/234	[14]	flat1000_60_0	60/60	[12,7,14,15,19,20,21]
dsjc500.9	126/?	[7,11,12,18,19,20,22,23]	dsjr500.1c	85/84	[12,14,19,20,22]	flat1000_76_0	82/76	[14,19,20,21,22,23]
dsjc1000.1	20/?	[6,7,11,14,19,20,21,22,23]	dsjr500.5	122/122	[14,19,20,22,23]	latin_square	97/?	[23]
dsjc1000.5	83/?	[6,11,14,19,20,21,22,23]	le450_25c	25/25	[12 ^p , ³⁵³ ,14,19,20,22,23]	C2000.5	146/?	[21 ¹]
dsjc1000.9	222/?	[21,22,23]	le450.25d	25/25	[12 ^{pp} , ³⁵³ ,14,19,20,22,23]	C4000.5	260/?	[21 ¹]

2 Conclusion

Traditional heuristics such as DSATUR and Tabucol remain strong baselines, consistently producing optimal or near-optimal colorings on DIMACS benchmark graphs. In many instances, Tabucol achieves the exact chromatic number, outperforming simple greedy methods.

More recent approaches based on Graph Neural Networks (GNNs), including models such as PI-GCN and PI-SAGE, demonstrate competitive performance, achieving low error rates and accurate color assignments on small and medium graphs. However, these models sometimes still lag behind well-tuned heuristic methods.

Other works propose hybrid or evolutionary algorithms, which combine global search with local optimization to improve solution quality and convergence speed. Additionally, deep learning approaches such as Fully Convolutional Networks (FCN) show promising results on large synthetic graph datasets, outperforming several classical ordering strategies.

Overall, while learning-based methods are becoming increasingly effective, metaheuristic and tabu-search based algorithms remain among the most reliable techniques for obtaining optimal graph colorings.